

ME101: Engineering Mechanics (3 1 0 8)

2023-2024 (II Semester)
Division II & IV

Instructor

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Area Moments of Inertia

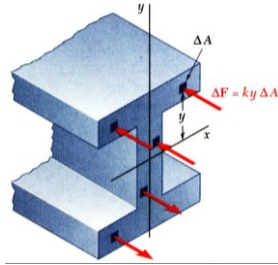
- Previously considered distributed forces which were proportional to the area or volume over which they act.
 - The resultant was obtained by summing or integrating over the areas or volumes.
 - The moment of the resultant about any axis was determined by computing the first moments of the areas or volumes about that axis.
- Will now consider forces which are proportional to the area or volume over which they act but also vary linearly with distance from a given axis.
 - the magnitude of the resultant depends on the first moment of the force distribution with respect to the axis.
 - The point of application of the resultant depends on the second moment of the distribution with respect to the axis.

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Area Moments of Inertia

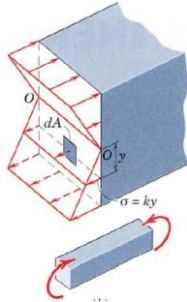


- Consider distributed forces $\Delta \vec{F}$ whose magnitudes are proportional to the elemental areas ΔA on which they act and also vary linearly with the distance of ΔA from a given axis.
- Example: Consider a beam subjected to pure bending. Internal forces vary linearly with distance from the neutral axis which passes through the section centroid.

$$\Delta \vec{F} = ky\Delta A$$

$$R = k \int y dA = 0 \quad \int y dA = Q_x = \text{first moment}$$

$$M = k \int y^2 dA \quad \int y^2 dA = \text{second moment}$$



First Moment of the whole section about the x-axis = $\bar{y}A = 0$ since the centroid of the section lies on the x-axis.

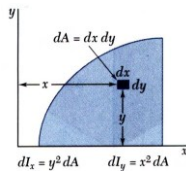
Second Moment or the Moment of Inertia of the beam section about x-axis is denoted by I_x and has units of $(\text{length})^4$ (never -ve)

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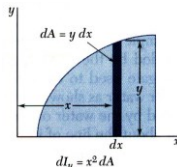
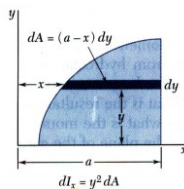
Area Moments of Inertia by Integration



- Second moments or moments of inertia of an area with respect to the x and y axes,

$$I_x = \int y^2 dA \quad I_y = \int x^2 dA$$

- Evaluation of the integrals is simplified by choosing dA to be a thin strip parallel to one of the coordinate axes.

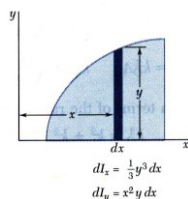
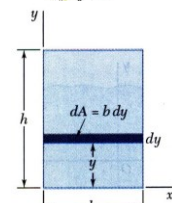


- For a rectangular area,

$$I_x = \int y^2 dA = \int_0^h y^2 b dy = \frac{1}{3} b h^3$$

- The formula for rectangular areas may also be applied to strips parallel to the axes,

$$dI_x = \frac{1}{3} y^3 dx \quad dI_y = x^2 dA = x^2 y dx$$



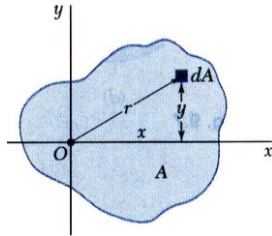
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Area Moments of Inertia

Polar Moment of Inertia



- The *polar moment of inertia* is an important parameter in problems involving torsion of cylindrical shafts and rotations of slabs.

$$J_0 = I_z = \int r^2 dA$$

- The polar moment of inertia is related to the rectangular moments of inertia,

$$J_0 = I_z = \int r^2 dA = \int (x^2 + y^2) dA = \int x^2 dA + \int y^2 dA = I_y + I_x$$

Moment of Inertia of an area is purely a mathematical property of the area and in itself has no physical significance.

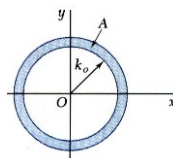
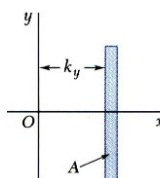
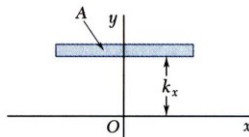
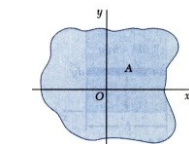
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Area Moments of Inertia

Radius of Gyration of an Area



- Consider area A with moment of inertia I_x . Imagine that the area is concentrated in a thin strip parallel to the x axis with equivalent I_x .

$$I_x = k_x^2 A \quad k_x = \sqrt{\frac{I_x}{A}}$$

$k_x =$ radius of gyration with respect to the x axis

- Similarly,

$$I_y = k_y^2 A \quad k_y = \sqrt{\frac{I_y}{A}}$$

$$J_0 = I_z = k_o^2 A = k_z^2 A \quad k_o = k_z = \sqrt{\frac{J_0}{A}}$$

$$k_o^2 = k_z^2 = k_x^2 + k_y^2$$

Radius of Gyration, k is a measure of distribution of area from a reference axis
Radius of Gyration is different from centroidal distances

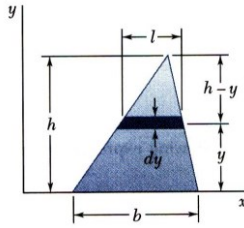
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Area Moments of Inertia

Example: Determine the moment of inertia of a triangle with respect to its base.



SOLUTION:

- A differential strip parallel to the x axis is chosen for dA .

$$dI_x = y^2 dA \quad dA = l dy$$

- For similar triangles,

$$\frac{l}{b} = \frac{h-y}{h} \quad l = b \frac{h-y}{h} \quad dA = b \frac{h-y}{h} dy$$

- Integrating dI_x from $y = 0$ to $y = h$,

$$I_x = \int y^2 dA = \int_0^h y^2 b \frac{h-y}{h} dy = \frac{b}{h} \int_0^h (hy^2 - y^3) dy$$

$$= \frac{b}{h} \left[h \frac{y^3}{3} - \frac{y^4}{4} \right]_0^h \quad I_x = \frac{bh^3}{12}$$

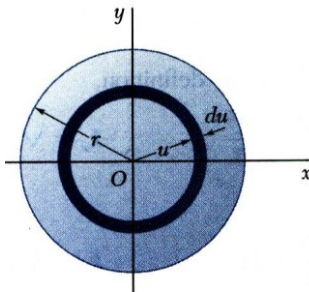
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Area Moments of Inertia

Example: a) Determine the centroidal polar moment of inertia of a circular area by direct integration.
b) Using the result of part a, determine the moment of inertia of a circular area with respect to a diameter.



- An annular differential area element is chosen,

$$dJ_O = u^2 dA \quad dA = 2\pi u du$$

$$J_O = \int dJ_O = \int_0^r u^2 (2\pi u du) = 2\pi \int_0^r u^3 du$$

$$J_O = \frac{\pi}{2} r^4$$

- From symmetry, $I_x = I_y$,

$$J_O = I_x + I_y = 2I_x \quad \frac{\pi}{2} r^4 = 2I_x$$

$$I_{\text{diameter}} = I_x = \frac{\pi}{4} r^4$$

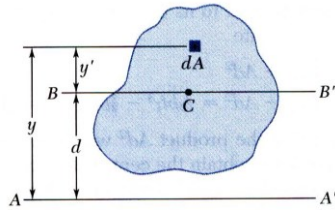
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Area Moments of Inertia

Parallel Axis Theorem



Parallel Axis theorem:

MI @ any axis =

MI @ centroidal axis + Ad^2

The two axes should be parallel to each other.

- Consider moment of inertia I of an area A with respect to the axis AA'

$$I = \int y^2 dA$$

- The axis BB' passes through the area centroid and is called a *centroidal axis*.

$$\begin{aligned} I &= \int y^2 dA = \int (y' + d)^2 dA \\ &= \int y'^2 dA + 2d \int y' dA + d^2 \int dA \end{aligned}$$

- Second term = 0 since centroid lies on BB' ($\int y' dA = y_c A$, and $y_c = 0$)

$$I = \bar{I} + Ad^2 \quad \text{Parallel Axis theorem}$$

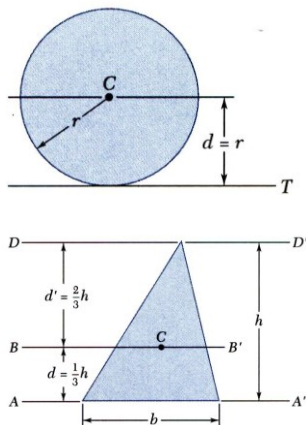
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Area Moments of Inertia

Parallel Axis Theorem



- Moment of inertia I_T of a circular area with respect to a tangent to the circle,

$$\begin{aligned} I_T &= \bar{I} + Ad^2 = \frac{1}{4}\pi r^4 + (\pi r^2)r^2 \\ &= \frac{5}{4}\pi r^4 \end{aligned}$$

- Moment of inertia of a triangle with respect to a centroidal axis,

$$\begin{aligned} I_{AA'} &= \bar{I}_{BB'} + Ad^2 \\ I_{BB'} &= I_{AA'} - Ad^2 = \frac{1}{12}bh^3 - \frac{1}{2}bh\left(\frac{1}{3}h\right)^2 \\ &= \frac{1}{36}bh^3 \end{aligned}$$

- The moment of inertia of a composite area A about a given axis is obtained by adding the moments of inertia of the component areas $A_1, A_2, A_3, \dots$, with respect to the same axis.

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Area Moments of Inertia: Standard MIs

	$\bar{I}_{x'} = \frac{1}{12}bh^3$ $\bar{I}_{y'} = \frac{1}{12}b^3h$ $I_x = \frac{1}{3}bh^3$ $I_y = \frac{1}{3}b^3h$ $J_C = \frac{1}{12}bh(b^2 + h^2)$		$I_x = I_y = \frac{1}{8}\pi r^4$ $J_O = \frac{1}{4}\pi r^4$
	$\bar{I}_{x'} = \frac{1}{36}bh^3$ $I_x = \frac{1}{12}bh^3$		$I_x = I_y = \frac{1}{16}\pi r^4$ $J_O = \frac{1}{8}\pi r^4$
	$\bar{I}_x = \bar{I}_y = \frac{1}{4}\pi r^4$ $J_O = \frac{1}{2}\pi r^4$		$\bar{I}_x = \frac{1}{4}\pi ab^3$ $\bar{I}_y = \frac{1}{4}\pi a^3b$ $J_O = \frac{1}{4}\pi ab(a^2 + b^2)$

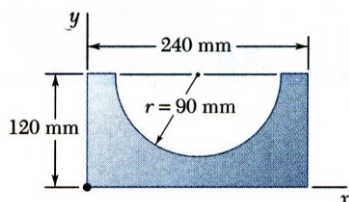
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Area Moments of Inertia

Example:

Determine the moment of inertia of the shaded area with respect to the x axis.

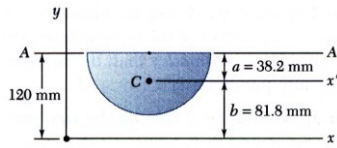


SOLUTION:

- Compute the moments of inertia of the bounding rectangle and half-circle with respect to the x axis.
- The moment of inertia of the shaded area is obtained by subtracting the moment of inertia of the half-circle from the moment of inertia of the rectangle.

Area Moments of Inertia

Example: Solution



$$a = \frac{4r}{3\pi} = \frac{(4)(90)}{3\pi} = 38.2 \text{ mm}$$

$$b = 120 - a = 81.8 \text{ mm}$$

$$A = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(90)^2$$

$$= 12.72 \times 10^3 \text{ mm}^2$$

SOLUTION:

- Compute the moments of inertia of the bounding rectangle and half-circle with respect to the x axis.

Rectangle:

$$I_x = \frac{1}{3}bh^3 = \frac{1}{3}(240)(120)^3 = 138.2 \times 10^6 \text{ mm}^4$$

Half-circle:

moment of inertia with respect to AA' ,

$$I_{AA'} = \frac{1}{8}\pi r^4 = \frac{1}{8}\pi(90)^4 = 25.76 \times 10^6 \text{ mm}^4$$

Moment of inertia with respect to x' ,

$$\bar{I}_{x'} = I_{AA'} - Aa^2 = (25.76 \times 10^6) - (12.72 \times 10^3)(38.2)^2$$

$$= 7.20 \times 10^6 \text{ mm}^4$$

moment of inertia with respect to x ,

$$I_x = \bar{I}_{x'} + Ab^2 = 7.20 \times 10^6 + (12.72 \times 10^3)(81.8)^2$$

$$= 92.3 \times 10^6 \text{ mm}^4$$

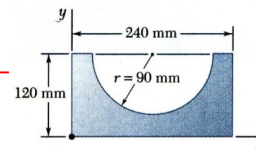
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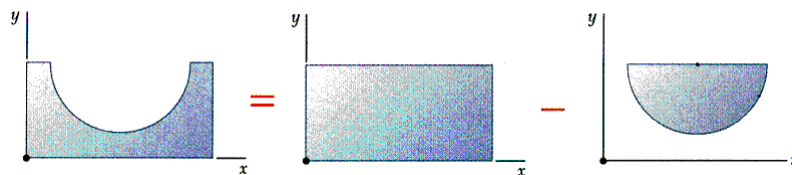
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Area Moments of Inertia

Example: Solution



- The moment of inertia of the shaded area is obtained by subtracting the moment of inertia of the half-circle from the moment of inertia of the rectangle.



$$I_x = 138.2 \times 10^6 \text{ mm}^4 - 92.3 \times 10^6 \text{ mm}^4$$

$$I_x = 45.9 \times 10^6 \text{ mm}^4$$

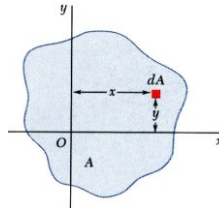
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Area Moments of Inertia

Products of Inertia: for problems involving unsymmetrical cross-sections and in calculation of MI about rotated axes. It may be +ve, -ve, or zero

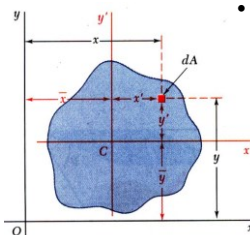


- Product of Inertia of area A w.r.t. x - y axes:

$$I_{xy} = \int xy dA$$

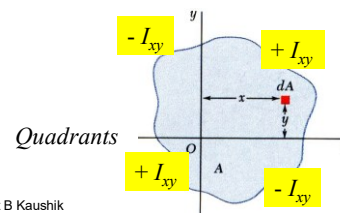
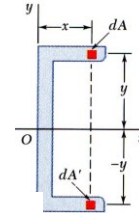
x and y are the coordinates of the element of area $dA = xy$

- When the x axis, the y axis, or both are an axis of symmetry, the product of inertia is zero.



- Parallel axis theorem for products of inertia:

$$I_{xy} = \bar{I}_{xy} + \bar{x}\bar{y}A$$



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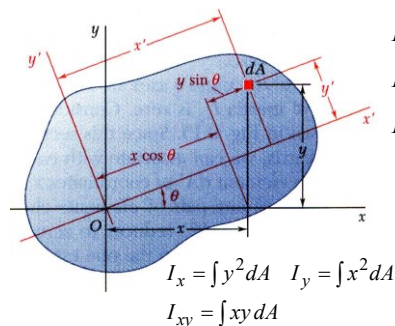
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Area Moments of Inertia

Rotation of Axes

Product of inertia is useful in calculating MI @ inclined axes.

→ Determination of axes about which the MI is a maximum and a minimum



$$I_{x'} = \int y'^2 dA = \int (y \cos \theta - x \sin \theta)^2 dA$$

$$I_{y'} = \int x'^2 dA = \int (x \cos \theta + y \sin \theta)^2 dA$$

$$I_{x'y'} = \int x' y' dA = \int (x \cos \theta + y \sin \theta)(y \cos \theta - x \sin \theta) dA$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2} \quad \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta \quad \cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

Moments and product of inertia w.r.t. new axes x' and y' ?

Note: $x' = x \cos \theta + y \sin \theta$

$$y' = y \cos \theta - x \sin \theta$$

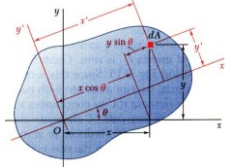
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Area Moments of Inertia

Rotation of Axes



$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

Adding first two eqns:

$$I_{x'} + I_{y'} = I_x + I_y = I_z \rightarrow \text{The Polar MI @ O}$$

Angle which makes $I_{x'}$ and $I_{y'}$ either max or min can be found by setting the derivative of either $I_{x'}$ or $I_{y'}$ w.r.t. θ equal to zero:

$$\frac{dI_{x'}}{d\theta} = (I_y - I_x) \sin 2\theta - 2I_{xy} \cos 2\theta = 0$$

Denoting this critical angle by α

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x}$$

- two values of 2α which differ by π since $\tan 2\alpha = \tan(2\alpha + \pi)$
- two solutions for α will differ by $\pi/2$
- one value of α will define the axis of maximum MI and the other defines the axis of minimum MI
- These two rectangular axes are called the principal axes of inertia

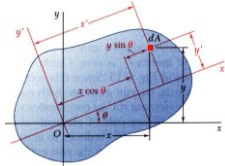
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Area Moments of Inertia

Rotation of Axes



$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x} \Rightarrow \sin 2\alpha = \cos 2\alpha \frac{2I_{xy}}{I_y - I_x}$$

Substituting in the third eqn for critical value of 2θ : $I_{x'y'} = 0$

→ Product of Inertia $I_{x'y'}$ is zero for the Principal Axes of inertia

Substituting $\sin 2\alpha$ and $\cos 2\alpha$ in first two eqns for **Principal Moments of Inertia**:

$$I_{\max} = \frac{I_x + I_y}{2} + \frac{1}{2} \sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{\min} = \frac{I_x + I_y}{2} - \frac{1}{2} \sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{xy@ \alpha} = 0$$

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Area Moments of Inertia

Mohr's Circle of Inertia: Following relations can be represented graphically by a diagram called Mohr's Circle

For given values of I_x , I_y , & I_{xy} , corresponding values of $I_{x'}$, $I_{y'}$, & $I_{x'y'}$ may be determined from the diagram for any desired angle θ .

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta \quad \tan 2\alpha = \frac{2I_{xy}}{I_y - I_x}$$

$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

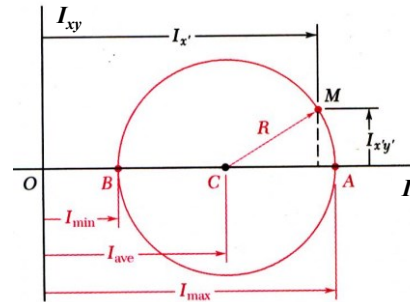
$$I_{\max} = \frac{I_x + I_y}{2} + \frac{1}{2} \sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{\min} = \frac{I_x + I_y}{2} - \frac{1}{2} \sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{xy@ \alpha} = 0$$

$$(I_{x'} - I_{ave})^2 + I_{x'y'}^2 = R^2$$

$$I_{ave} = \frac{I_x + I_y}{2} \quad R = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$



- At the points A and B, $I_{x'y'} = 0$ and $I_{x'}$ is a maximum and minimum, respectively.

$$I_{\max, \min} = I_{ave} \pm R$$

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Area Moments of Inertia

Mohr's Circle of Inertia: Construction

$$\tan 2\alpha = \frac{2I_{xy}}{I_y - I_x}$$

$$I_{x'} = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

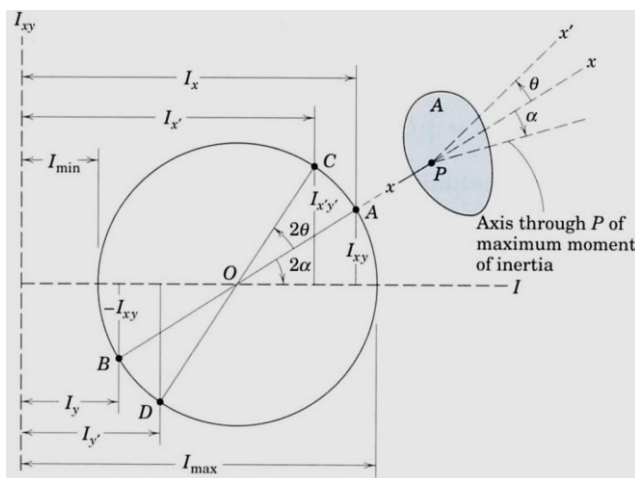
$$I_{y'} = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{x'y'} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

$$I_{\max} = \frac{I_x + I_y}{2} + \frac{1}{2} \sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{\min} = \frac{I_x + I_y}{2} - \frac{1}{2} \sqrt{(I_x - I_y)^2 + 4I_{xy}^2}$$

$$I_{xy@ \alpha} = 0$$



Choose horz axis $\rightarrow$ MI

Choose vert axis $\rightarrow$ PI

Point A – known $\{I_{x'}, I_{xy}\}$

Point B – known $\{I_y, -I_{xy}\}$

Circle with dia AB

Angle α for Area

$\rightarrow$ Angle 2α to horz (same sense) $\rightarrow I_{\max}, I_{\min}$

Angle x to $x' = \theta$

$\rightarrow$ Angle OA to OC = 2θ

$\rightarrow$ Same sense

Point C $\rightarrow I_{x'}, I_{x'y'}$

Point D $\rightarrow I_{y'}$

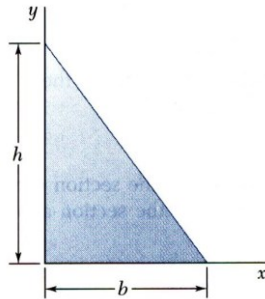
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Area Moments of Inertia

Example: Product of Inertia



SOLUTION:

- Determine the product of inertia using direct integration with the parallel axis theorem on vertical differential area strips
- Apply the parallel axis theorem to evaluate the product of inertia with respect to the centroidal axes.

Determine the product of inertia of the right triangle (a) with respect to the x and y axes and (b) with respect to centroidal axes parallel to the x and y axes.

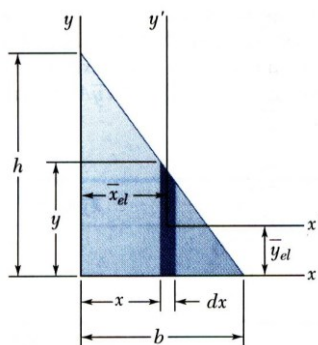
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Area Moments of Inertia

Examples



SOLUTION:

- Determine the product of inertia using direct integration with the parallel axis theorem on vertical differential area strips

$$y = h \left(1 - \frac{x}{b} \right) \quad dA = y dx = h \left(1 - \frac{x}{b} \right) dx$$

$$\bar{x}_{el} = x \quad \bar{y}_{el} = \frac{1}{2} y = \frac{1}{2} h \left(1 - \frac{x}{b} \right)$$

Integrating dI_x from $x = 0$ to $x = b$,

$$\begin{aligned} I_{xy} &= \int dI_{xy} = \int \bar{x}_{el} \bar{y}_{el} dA = \int_0^b x \left(\frac{1}{2} h \right) \left(1 - \frac{x}{b} \right)^2 dx \\ &= h^2 \int_0^b \left(\frac{x}{2} - \frac{x^2}{b} + \frac{x^3}{2b^2} \right) dx = h^2 \left[\frac{x^2}{4} - \frac{x^3}{3b} + \frac{x^4}{8b^2} \right]_0^b \end{aligned}$$

$$I_{xy} = \frac{1}{24} b^2 h^2$$

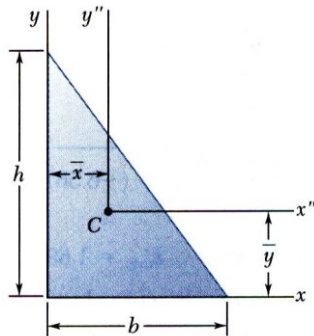
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Area Moments of Inertia

Examples



SOLUTION

- Apply the parallel axis theorem to evaluate the product of inertia with respect to the centroidal axes.

$$\bar{x} = \frac{1}{3}b \quad \bar{y} = \frac{1}{3}h$$

With the results from part a,

$$I_{xy} = \bar{I}_{x''y''} + \bar{x}\bar{y}A$$

$$\bar{I}_{x''y''} = \frac{1}{24}b^2h^2 - \left(\frac{1}{3}b\right)\left(\frac{1}{3}h\right)\left(\frac{1}{2}bh\right)$$

$$\bar{I}_{x''y''} = -\frac{1}{72}b^2h^2$$

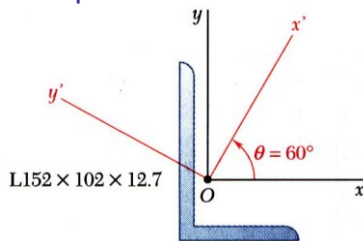
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Area Moments of Inertia

Example: Mohr's Circle of Inertia



The moments and product of inertia with respect to the x and y axes are $I_x = 7.24 \times 10^6 \text{ mm}^4$, $I_y = 2.61 \times 10^6 \text{ mm}^4$, and $I_{xy} = -2.54 \times 10^6 \text{ mm}^4$.

Using Mohr's circle, determine (a) the principal axes about O , (b) the values of the principal moments about O , and (c) the values of the moments and product of inertia about the x' and y' axes

SOLUTION:

- Plot the points (I_x, I_{xy}) and $(I_y, -I_{xy})$. Construct Mohr's circle based on the circle diameter between the points.
- Based on the circle, determine the orientation of the principal axes and the principal moments of inertia.
- Based on the circle, evaluate the moments and product of inertia with respect to the $x'y'$ axes.

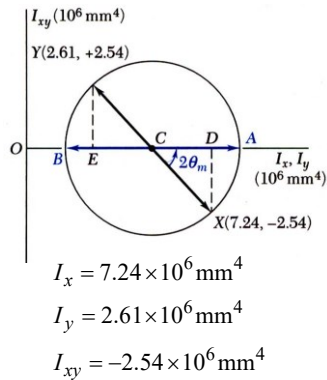
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Area Moments of Inertia

Example: Mohr's Circle of Inertia



SOLUTION:

- Plot the points (I_x, I_{xy}) and $(I_y, -I_{xy})$. Construct Mohr's circle based on the circle diameter between the points.

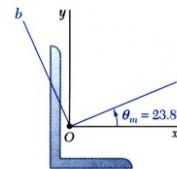
$$OC = I_{ave} = \frac{1}{2}(I_x + I_y) = 4.925 \times 10^6 \text{ mm}^4$$

$$CD = \frac{1}{2}(I_x - I_y) = 2.315 \times 10^6 \text{ mm}^4$$

$$R = \sqrt{(CD)^2 + (DX)^2} = 3.437 \times 10^6 \text{ mm}^4$$

- Based on the circle, determine the orientation of the principal axes and the principal moments of inertia.

$$\tan 2\theta_m = \frac{DX}{CD} = 1.097 \quad 2\theta_m = 47.6^\circ \quad \theta_m = 23.8^\circ$$



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Area Moments of Inertia

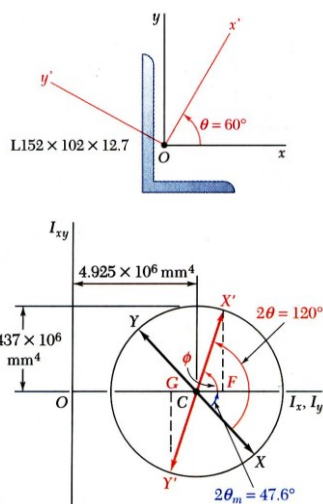
Example: Mohr's Circle of Inertia

$$OC = I_{ave} = 4.925 \times 10^6 \text{ mm}^4$$

$$R = 3.437 \times 10^6 \text{ mm}^4$$

- Based on the circle, evaluate the moments and product of inertia with respect to the $x'y'$ axes.

The points X' and Y' corresponding to the x' and y' axes are obtained by rotating CX and CY counterclockwise through an angle $\theta = 2(60^\circ) = 120^\circ$. The angle that CX' forms with the horz is $\phi = 120^\circ - 47.6^\circ = 72.4^\circ$.



$$I_{x'} = OF = OC + CX' \cos \phi = I_{ave} + R \cos 72.4^\circ$$

$$I_{x'} = 5.96 \times 10^6 \text{ mm}^4$$

$$I_{y'} = OG = OC - CY' \cos \phi = I_{ave} - R \cos 72.4^\circ$$

$$I_{y'} = 3.89 \times 10^6 \text{ mm}^4$$

$$I_{x'y'} = FX' = CY' \sin \phi = R \sin 72.4^\circ$$

$$I_{x'y'} = 3.28 \times 10^6 \text{ mm}^4$$

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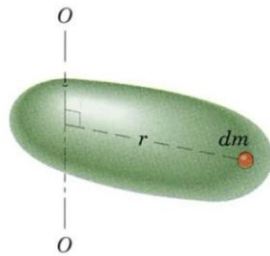
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Area Moments of Inertia

Mass Moments of Inertia (I): Important in Rigid Body Dynamics

- I is a measure of distribution of mass of a rigid body w.r.t. the axis in question (constant property for that axis).
- Units are (mass)(length)² $\rightarrow$ kg.m²



Consider a three dimensional body of mass m
Mass moment of inertia of this body about axis O-O:

$$I = \int r^2 dm$$

Integration is over the entire body.

r = perpendicular distance of the mass element dm from the axis O-O

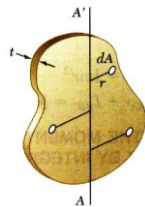
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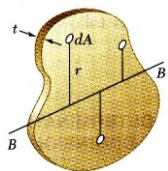
Area Moments of Inertia

Moments of Inertia of Thin Plates



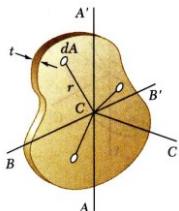
- For a thin plate of uniform thickness t and homogeneous material of density ρ , the mass moment of inertia with respect to axis AA' contained in the plate is

$$\begin{aligned} I_{AA'} &= \int r^2 dm = \rho t \int r^2 dA \\ &= \rho t I_{AA',area} \end{aligned}$$



- Similarly, for perpendicular axis BB' which is also contained in the plate,

$$I_{BB'} = \rho t I_{BB',area}$$



- For the axis CC' which is perpendicular to the plate,

$$\begin{aligned} I_{CC'} &= \rho t J_{C,area} = \rho t (I_{AA',area} + I_{BB',area}) \\ &= I_{AA'} + I_{BB'} \end{aligned}$$

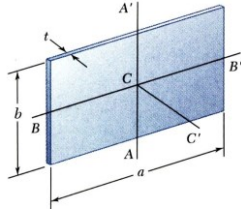
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Area Moments of Inertia

Moments of Inertia of Thin Plates

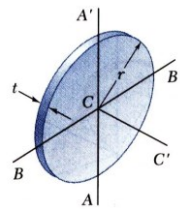


- For the principal centroidal axes on a rectangular plate,

$$I_{AA'} = \rho t I_{AA',area} = \rho t \left(\frac{1}{12} a^3 b \right) = \frac{1}{12} m a^2$$

$$I_{BB'} = \rho t I_{BB',area} = \rho t \left(\frac{1}{12} a b^3 \right) = \frac{1}{12} m b^2$$

$$I_{CC'} = I_{AA',mass} + I_{BB',mass} = \frac{1}{12} m (a^2 + b^2)$$



- For centroidal axes on a circular plate,

$$I_{AA'} = I_{BB'} = \rho t I_{AA',area} = \rho t \left(\frac{1}{4} \pi r^4 \right) = \frac{1}{4} m r^2$$

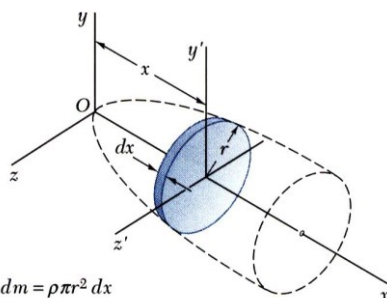
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Area Moments of Inertia

Moments of Inertia of a 3D Body by Integration



$$dm = \rho \pi r^2 dx$$

$$dI_x = \frac{1}{2} r^2 dm$$

$$dI_y = dI_{y'} + x^2 dm = \left(\frac{1}{4} r^2 + x^2 \right) dm$$

$$dI_z = dI_{z'} + x^2 dm = \left(\frac{1}{4} r^2 + x^2 \right) dm$$

- Moment of inertia of a homogeneous body is obtained from double or triple integrations of the form

$$I = \rho \int r^2 dV$$

- For bodies with two planes of symmetry, the moment of inertia may be obtained from a single integration by choosing thin slabs perpendicular to the planes of symmetry for dm .

- The moment of inertia with respect to a particular axis for a composite body may be obtained by adding the moments of inertia with respect to the same axis of the components.

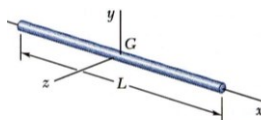
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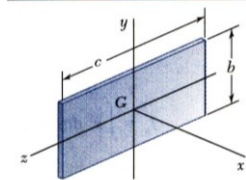
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Area Moments of Inertia

MI of some common geometric shapes



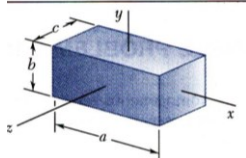
$$I_y = I_z = \frac{1}{12} mL^2$$



$$I_x = \frac{1}{12} m(b^2 + c^2)$$

$$I_y = \frac{1}{12} mc^2$$

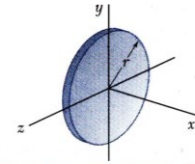
$$I_z = \frac{1}{12} mb^2$$



$$I_x = \frac{1}{12} m(b^2 + c^2)$$

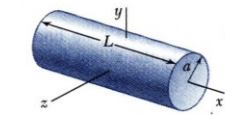
$$I_y = \frac{1}{12} m(c^2 + a^2)$$

$$I_z = \frac{1}{12} m(a^2 + b^2)$$



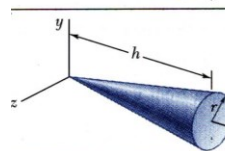
$$I_x = \frac{1}{2} mr^2$$

$$I_y = I_z = \frac{1}{4} mr^2$$



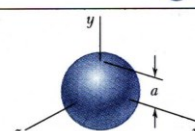
$$I_x = \frac{1}{2} ma^2$$

$$I_y = I_z = \frac{1}{12} m(3a^2 + L^2)$$



$$I_x = \frac{3}{10} ma^2$$

$$I_y = I_z = \frac{3}{5} m(\frac{1}{4} a^2 + h^2)$$



$$I_x = I_y = I_z = \frac{2}{5} ma^2$$

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Mass Moment of Inertia

Plane Kinetics of Rigid Bodies: **Force, Mass, & Acceleration**

- **Application to three cases of rigid body motion:**

- **Translation, Fixed Axis Rotation, and General Plane Motion**

Fixed Axis Rotation: Mass moment of inertia will be extensively used.

Mass Moments of Inertia

- Required in the analysis of any body that has rotational acceleration @ an axis
- Mass m of a body is a measure of resistance to translational acceleration
- Area moment of inertia is a measure of the distribution of area @ the axis
- **Mass Moment of Inertia I is a measure of resistance to rotational acceleration of the body**

Mass moment of inertia of the body @ O-O:

$$I = \int r^2 dm$$

$$I = \sum r_i^2 m_i$$

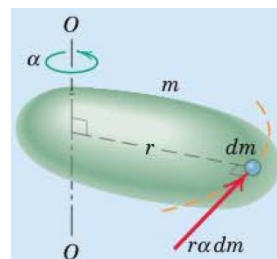
$$I = \rho \int r^2 dV$$

If mass density ρ is constant throughout the body

Units of Mass moment of inertia: kg m^2

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Mass Moment of Inertia

Mass Moments of Inertia

Radius of Gyration (k)

- about an axis for which I is defined:

$$k = \sqrt{\frac{I}{m}} \quad \text{or} \quad I = k^2 m$$

Parallel Axis Theorem (Transfer of Axes)

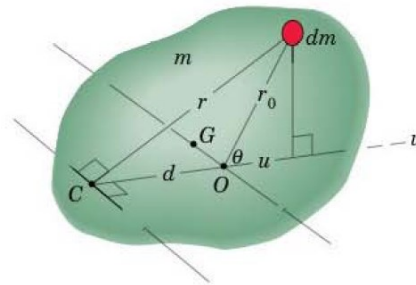
- Mass moment of inertia about any axis parallel to the axis passing through mass center G :

$$I = \bar{I} + md^2$$

- Radius of gyration @ an axis through C

$$k^2 = \bar{k}^2 + d^2$$

$\bar{I}$ and $\bar{k}$ are the values @ an axis through mass center



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Mass Moment of Inertia

Mass Moments of Inertia

Plane Motion:

Mass MI of the plate (with motion in x-y plane)
@ z-axis through O:

$$I_O = \int r^2 dm$$

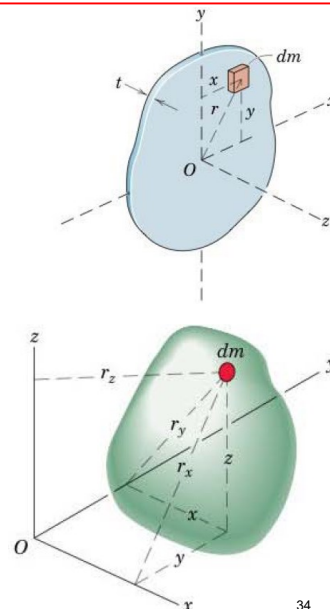
$$I_O = \sum r_i^2 m_i$$

3-D Motion:

$$I_{xx} = \int r_x^2 dm = \int (y^2 + z^2) dm$$

$$I_{yy} = \int r_y^2 dm = \int (z^2 + x^2) dm$$

$$I_{zz} = \int r_z^2 dm = \int (x^2 + y^2) dm$$



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Mass Moment of Inertia

Mass Moments of Inertia

Thin Plates

Relationship between mass moments of inertia and area moments of inertia exists in case of **flat plates**.

t = constant thickness of the plate,

ρ = constant mass density of the plate

Mass MI I_{zz} of the plate @ z-axis normal to the plate:

$$I_{zz} = \int r^2 dm = \rho t \int r^2 dA = \rho t I_z$$

Mass MI @ z-axis = mass per unit area (ρt) x Polar MI of the plate area @ z-axis

If t is small compared to the dimensions of the plate in its plane:

Mass MI I_{xx} and I_{yy} of the plate @ x- and y-axes are closely approximated by:

$$I_{xx} = \int y^2 dm = \rho t \int y^2 dA = \rho t I_x$$

For thin plates

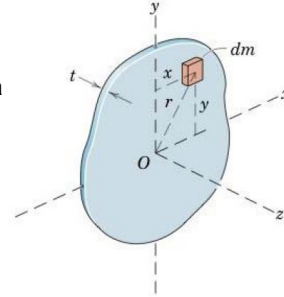
$$I_{yy} = \int x^2 dm = \rho t \int x^2 dA = \rho t I_y$$

$$I_{zz} = I_{xx} + I_{yy} \quad \text{For thin plates}$$

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Mass Moment of Inertia

Mass Moments of Inertia

Products of Inertia: used in the expression for angular momentum of rigid bodies under 3-D motion

$$I_{xy} = I_{yx} = \int xy dm$$

$$I_{xz} = I_{zx} = \int xz dm$$

$$I_{yz} = I_{zy} = \int yz dm$$

$$I_{xy} = \bar{I}_{xy} + m d_x d_y$$

$$I_{xz} = \bar{I}_{xz} + m d_x d_z$$

$$I_{yz} = \bar{I}_{yz} + m d_y d_z$$

Parallel Axis Theorem

→ extremely useful while determining mass MI @ any axis OM

Direction cosines of OM: l, m, n

Unit vector along OM: $\lambda = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$

$$I_M = \int h^2 dm = \int (\mathbf{r} \times \lambda) \cdot (\mathbf{r} \times \lambda) dm$$

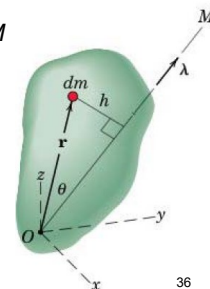
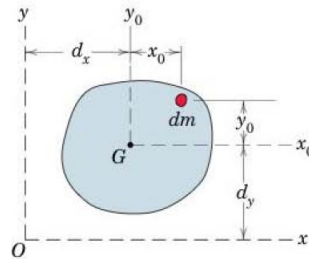
$$(\mathbf{r} \times \lambda) = (yn - zm)\mathbf{i} + (zl - xn)\mathbf{j} + (xm - yl)\mathbf{k}$$

$$I_M = I_{xx}l^2 + I_{yy}m^2 + I_{zz}n^2 - 2I_{xy}lm - 2I_{xz}ln - 2I_{yz}mn$$

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That is All!